

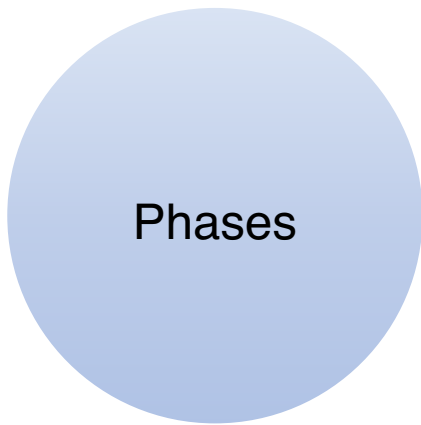
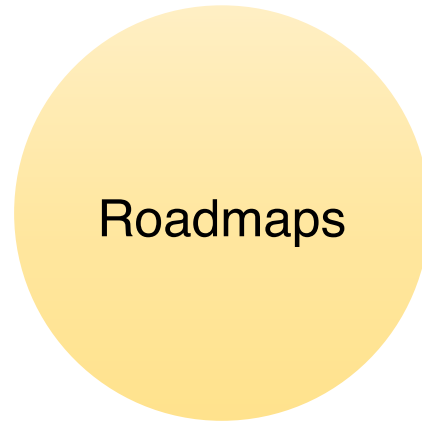
INFOSYS 704

IT Consultancy

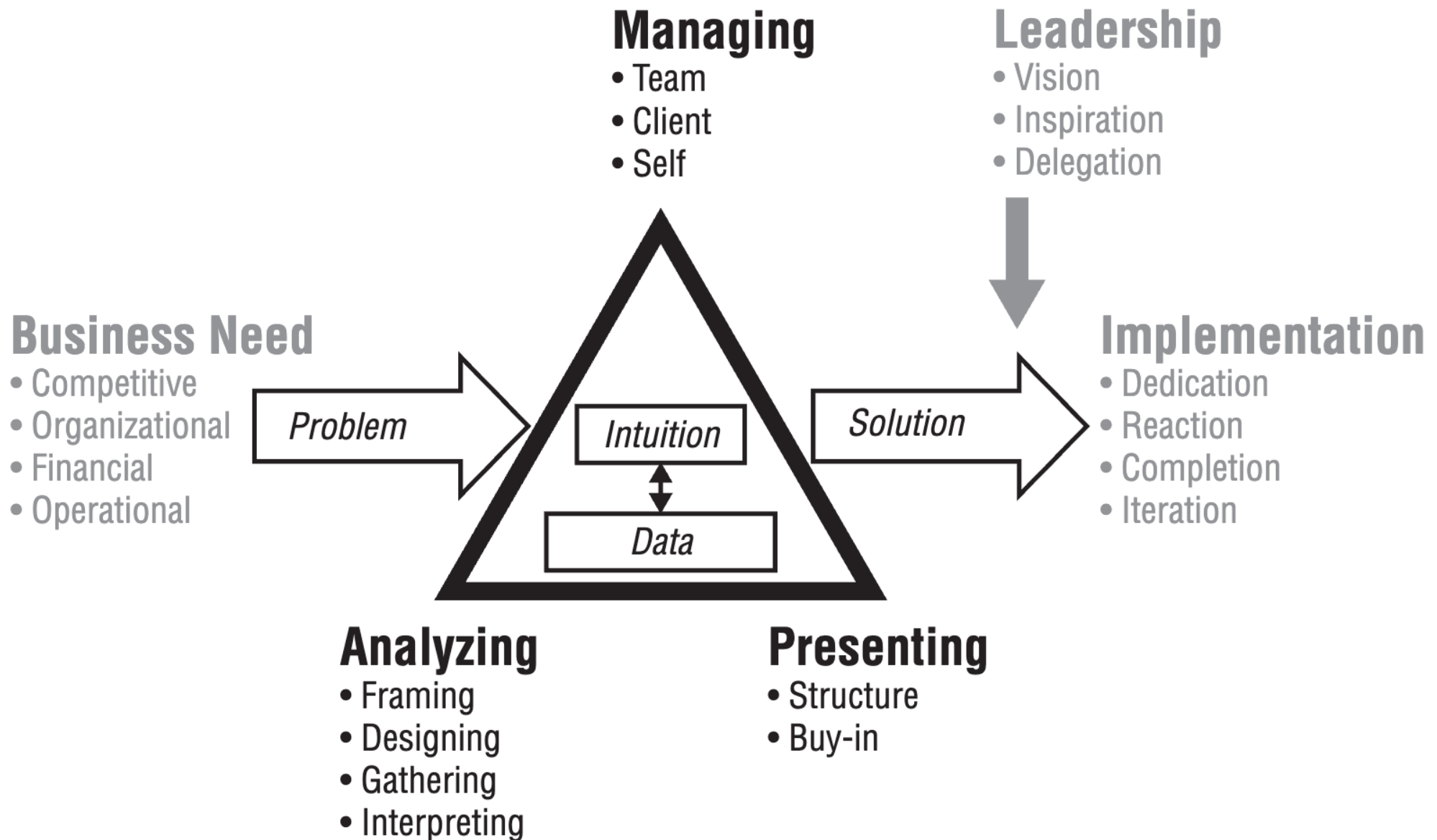
Lecture 7

Implementation Roadmaps





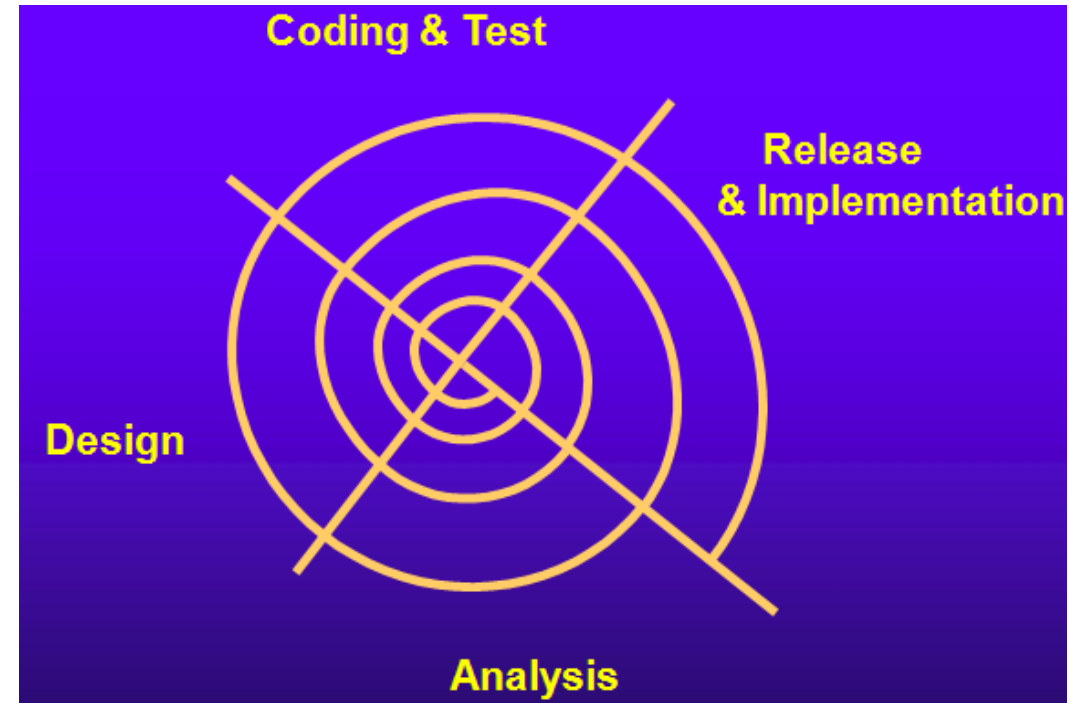
McKinsey Problem Solving Model



Implementation Roadmaps

Traditional Software Development Lifecycle Models

Waterfall Model
(Boehm, 1981)



Spiral Model
(Boehm, 1988)

Prototyping Model
(Pressman, 1997)

Rational Unified Process: The Iterative Model

(Rational, 2002)

Process Workflows

Business Modeling

Requirements

Analysis & Design

Implementation

Test

Deployment

Supporting Workflows

Configuration & Change Mgmt

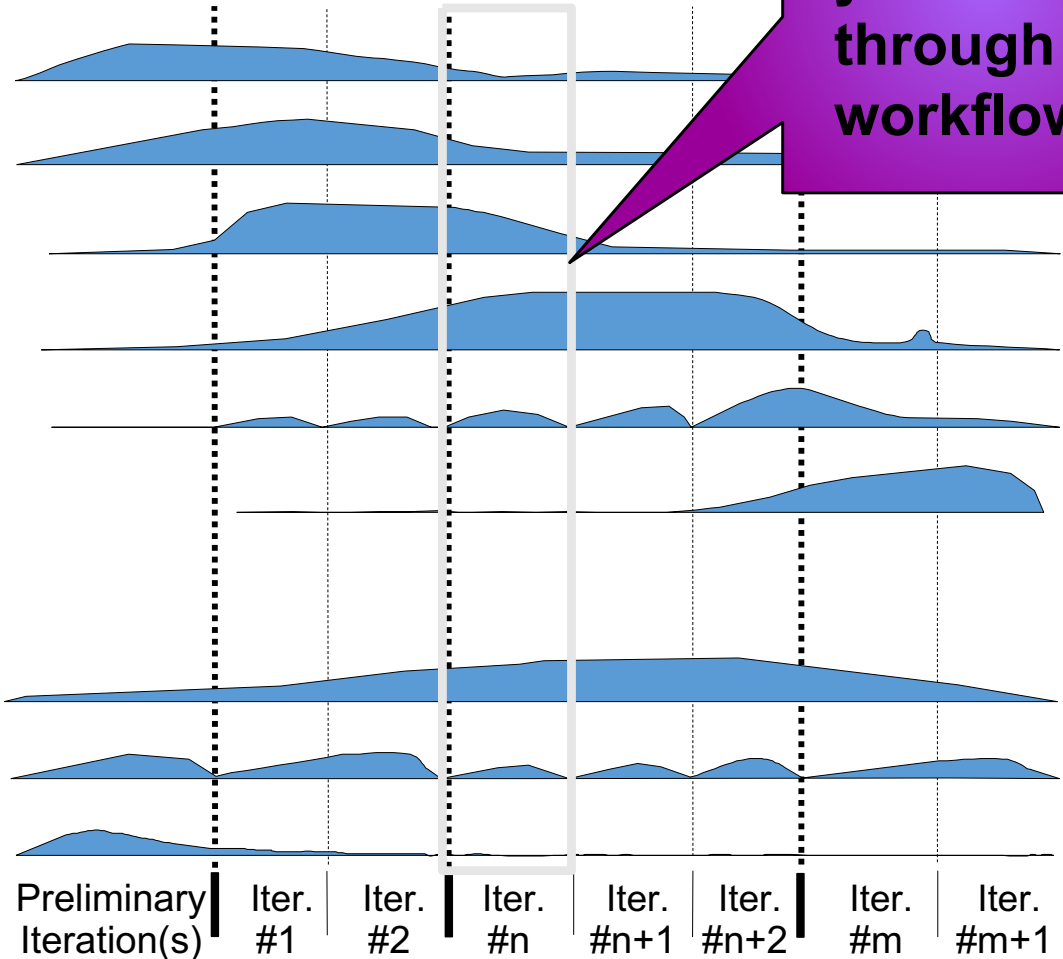
Project Management

Environment

Workflows
group
activities
logically

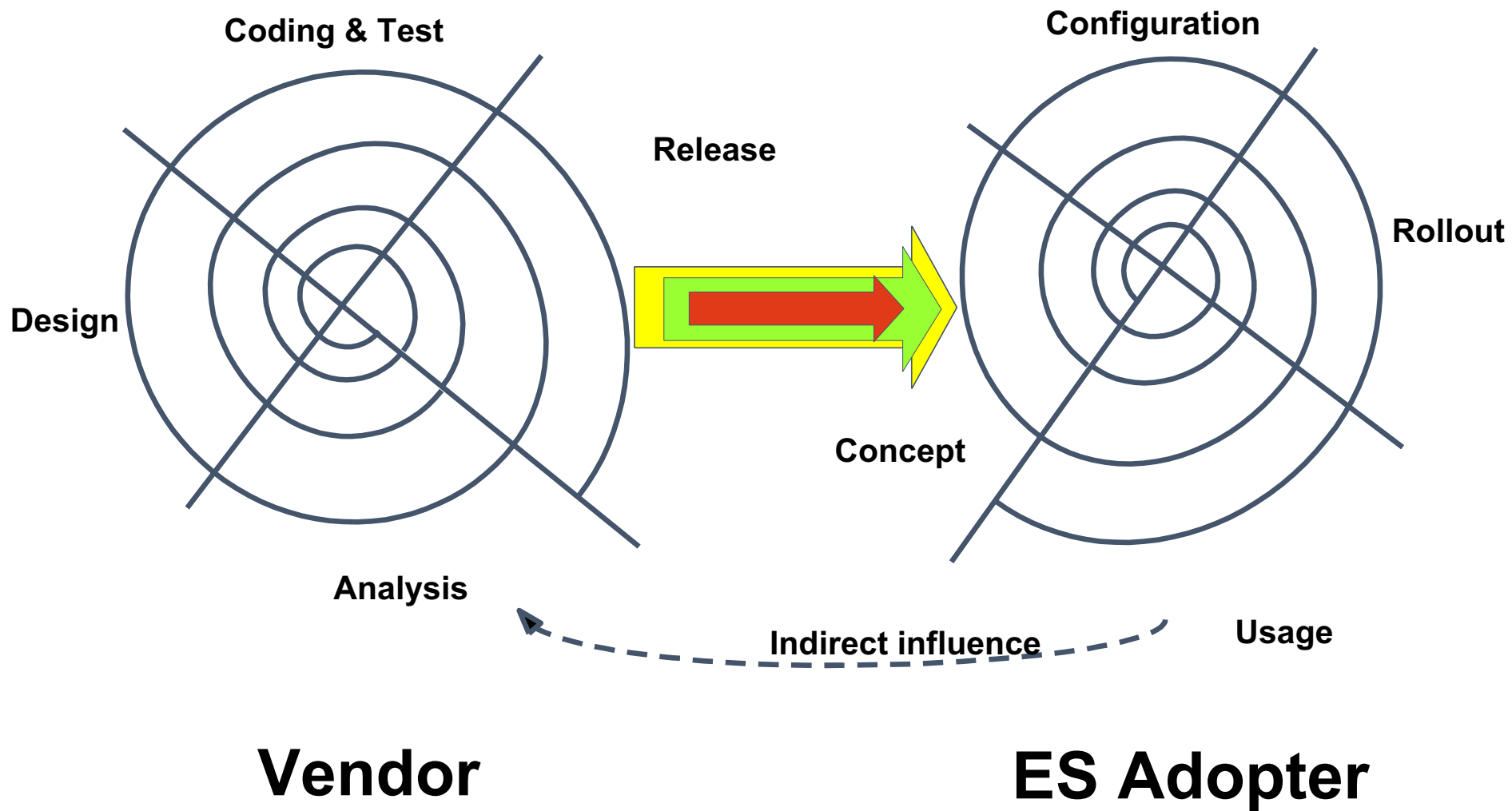
Phases

Inception Elaboration Construction



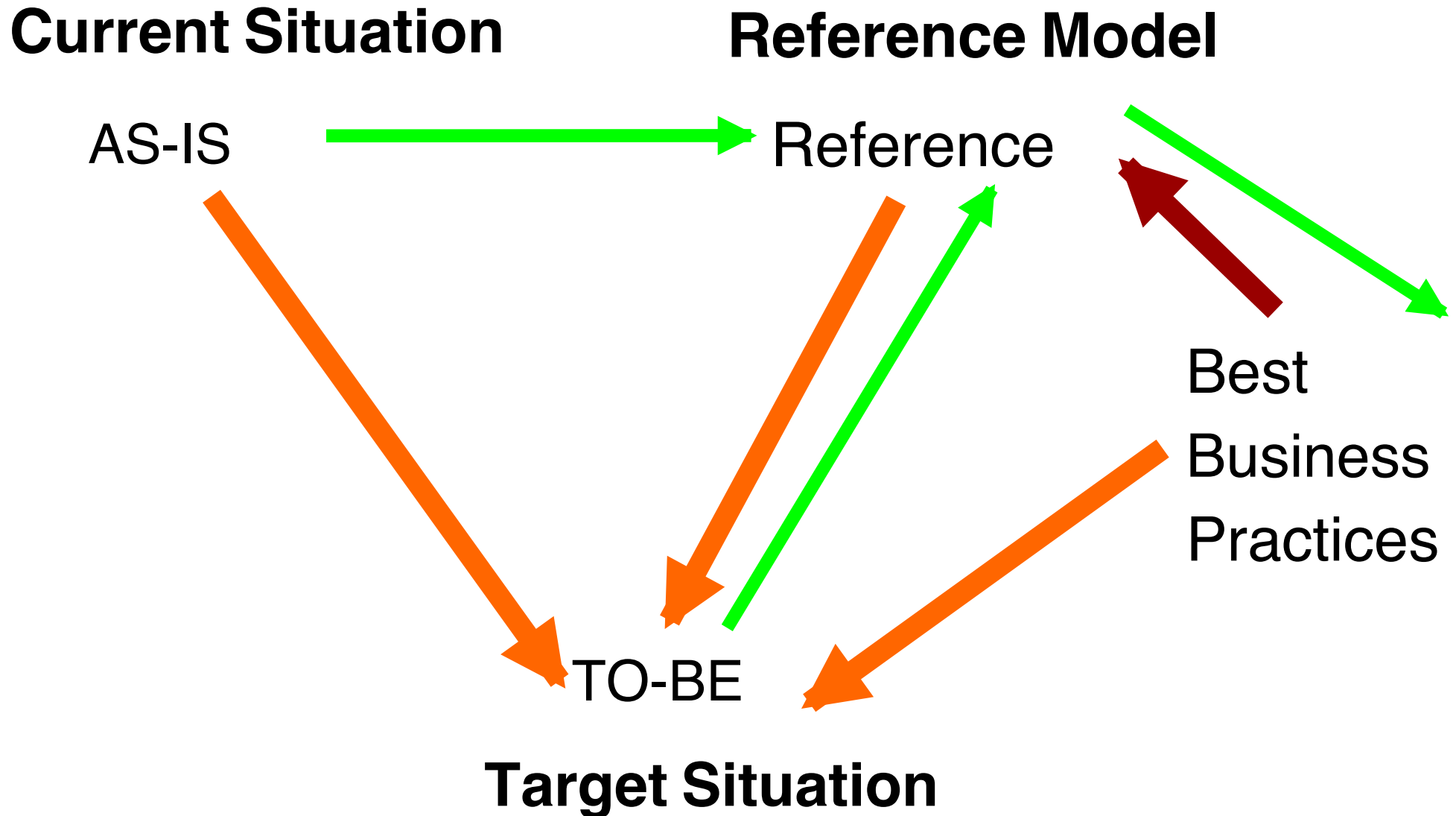
Iterations

ESs Divided Software Lifecycle

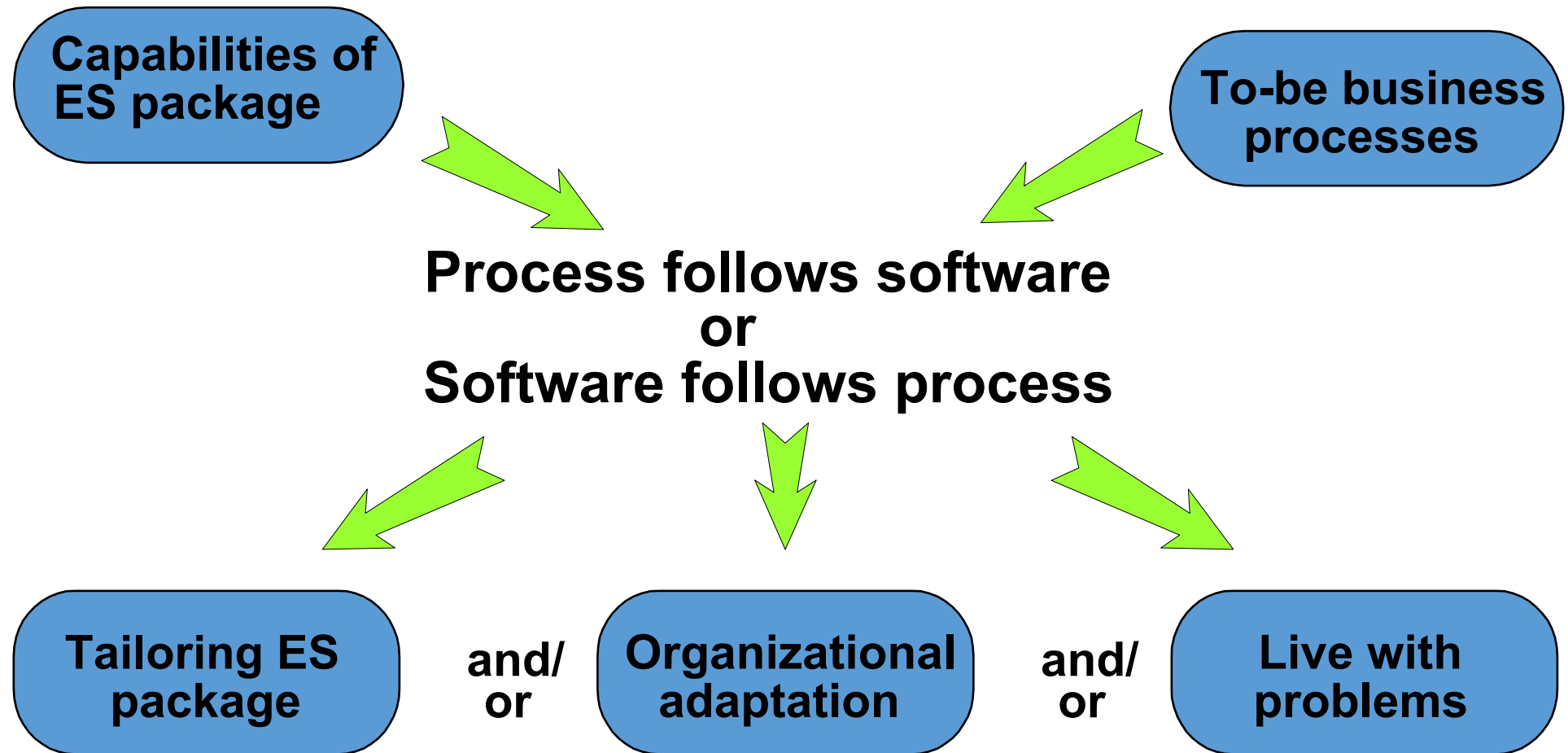


(Brehm & Markus, 2000)

AS-IS TO-BE BBP REFERENCE Models

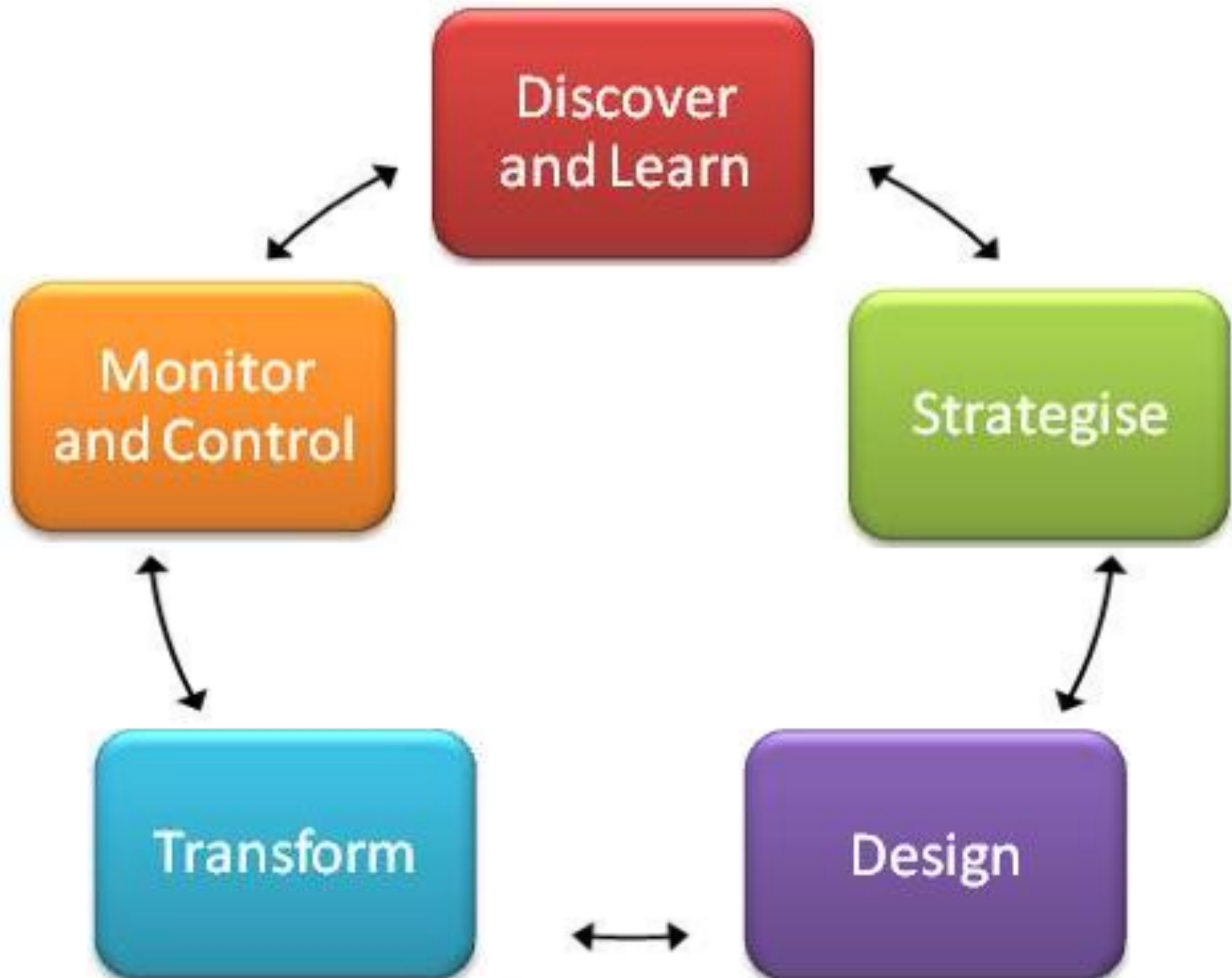


ES Implementation Process



- **Configuration**
- **Modification**

Sustainable Business Transformation Roadmap

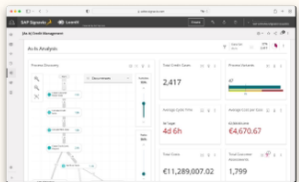


SAP Activate with Signavio and LeanIX

DISCOVER

Achieve business process and enterprise architecture transparency.

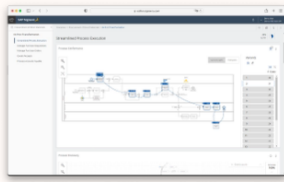
- Analyze as-is processes and process variants with SAP Process Insights and SAP Signavio Process Transformation Suite
- Review performance KPIs and identify improvements



PREPARE

Support business case creation and enable strategic decision-making.

- Define the scope of planned improvements using SAP Signavio Process Transformation's mining algorithms and direct connectors to SAP
- Analyze SAP ECC data to structure SAP S/4HANA-based processes, identify key KPIs and benchmarks



EXPLORE

Define target processes and target enterprise architecture.

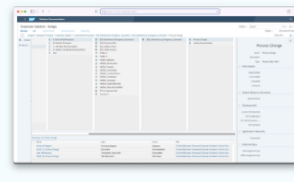
- Simulate target process performance using SAP Signavio Process Manager and SAP Best Practices
- Quickly uncover process deficiencies



REALIZE

Implement process improvements and define the transformation roadmap.

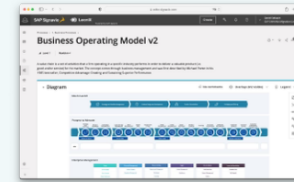
- Push approved business requirements for new processes to SAP Solution Manager, making them available to IT teams
- Start process implementation and testing



DEPLOY

Collaborate with stakeholders and track architecture transformation.

- Utilize the SAP Enable Now integration with SAP Signavio Process Collaboration Hub to drive adoption
- Supply stakeholders with onboarding materials



RUN

Continuously improve processes and enterprise architecture.

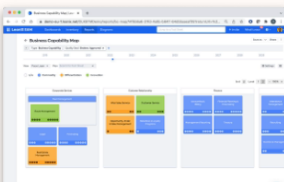
- Monitor performance of new processes within SAP Signavio Process Collaboration Hub
- Evaluate process conformance and leverage experience data for continuous process improvement



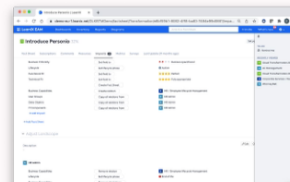
- Explore current EA through Application and IT Component Landscape Reports, and Interface Maps
- Identify dependencies between SAP and non-SAP applications
- Identify risks early



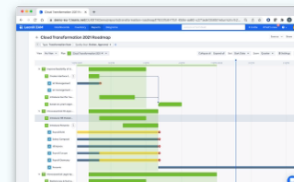
- Create configurable landscape reports (grouped by Business Capabilities, Processes, ITC, etc.)
- Map business capabilities to value drivers and support strategic decision making



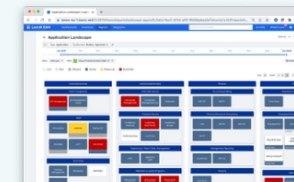
- Model target SAP S/4HANA architecture with Interface Maps, Data Flow Diagrams, IT Component and Application Landscape Reports
- Identify gaps between current and target architecture



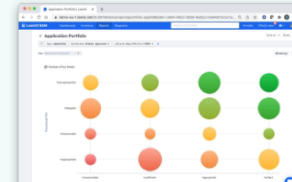
- Maintain IT landscape maps
- Visualize the impact of your transformation over time
- Perform timeline sequencing and scenario planning in LeanIX EAM



- Confirm completed architectural changes with teammates directly in LeanIX
- Instantly document modifications and communicate changes to your IT and business landscape



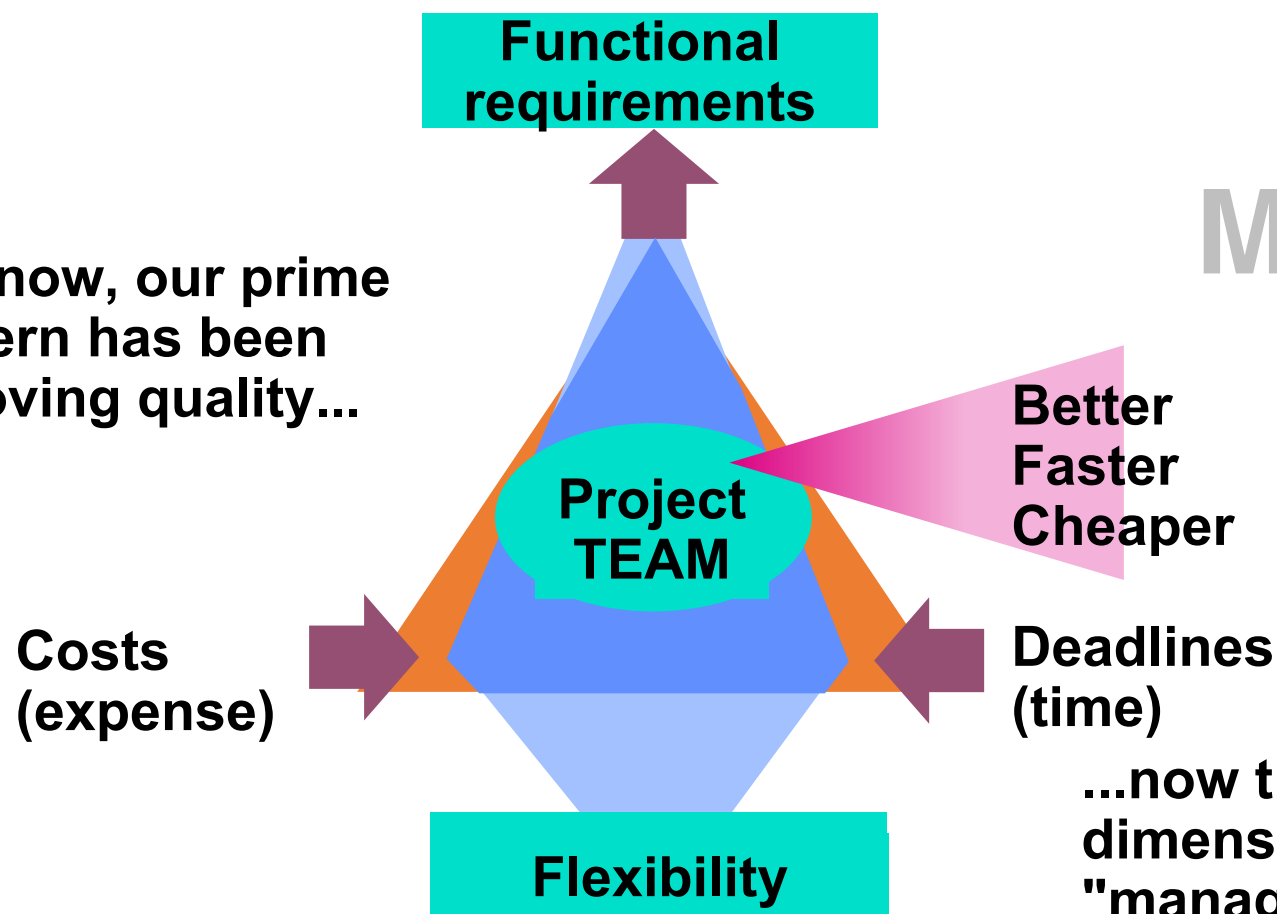
- Use EA insights to guide your improvement efforts
- Monitor technical and functional fit of all architectural elements by assessing the Application Portfolio Report



Project Management

Project Management Triangle

Until now, our prime concern has been improving quality...



...now there is a new dimension to consider - "managing the evolution"

The relationship between the aims must be right !

Aim: We must concentrate on optimizing the whole process, not just parts of it!

Davenport (2000) suggests thinking of ES implementation as a new business venture and/or a business change program not just a simple project. Stages with piecemeal capital allocation as long as it does not hinder integration, incentives, commn.,...

When IT Projects go wrong, they do so spectacularly

State computer project cost soars; Insurance system upgrades \$6.4 million over budget

The unemployment insurance computer project was supposed to cost \$17.2 million. That figure has been recalculated at \$23.6 million, or **37% more** than originally planned.

Patrick Marley, 2 April 2005, The Milwaukee Journal Sentinel

Audit finds federal project went \$64m over the Edge

There were constant tensions between FACS and Centrelink about how the project would be funded, with one dispute ending in Centrelink writing off \$15 million. The project's **costs doubled**, and at one stage it was **19 months behind schedule**.

Selina Mitchell, 19 April 2005, The Australian

LINZ \$35 million project blow-out may reach \$150M

THE cost of Land Information New Zealand's land title computerisation project, budgeted in 1997 for \$95 million, could escalate to **\$150 million**.

Amanda Wells, 21 June 1999, Infotech Weekly

Spies' IT system running 50 per cent over budget

Problems have cost several hundred million pounds, say MPs. A project to renew the UK Security Service's IT infrastructure has gone at least 50 per cent over budget, according to MPs.

Parliamentary Correspondent, 8 April 2005, VNUNet

The chances for success are dwindling

? Complex IT Projects that involved Enterprise Systems or complex solutions, often do not meet expectations

IT still not delivering for business

Project failure rates still at unacceptably high levels.

Nearly a third of managers experience IT **project failure rates of between 41 and 70 per cent**, according to research ... In a survey of 300 executives, 35 per cent of business managers and 27 per cent of IT managers admitted to IT project failures between these levels.

Steve Ranger 21 February 2005 VNUnet Newswire

Chaos Is Back

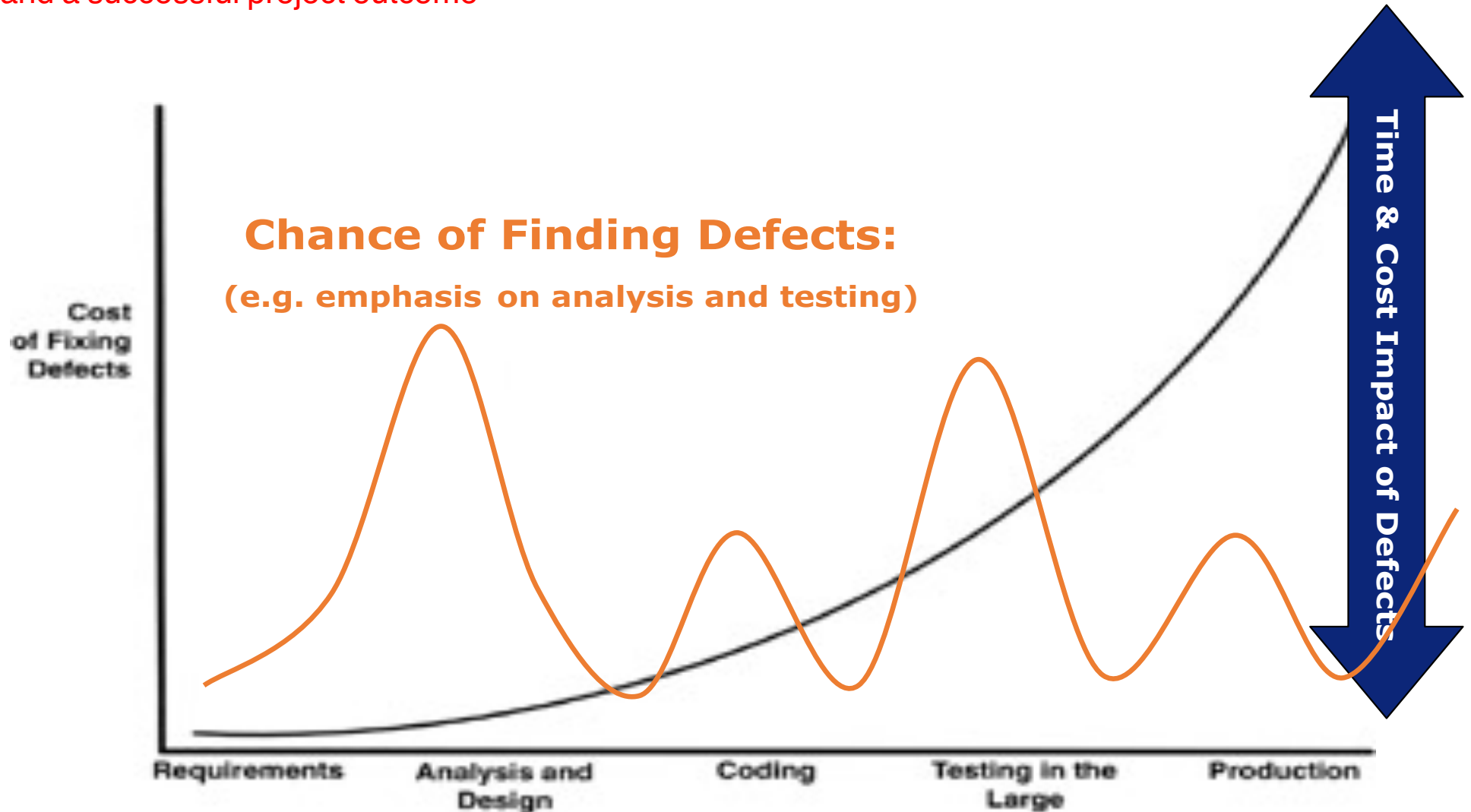
We're losing ground. **Only 28% of IT projects** succeed these days, down from 34% a year or two ago. Outright **failures** - IT projects canceled before completion - are up to 18% from 15%. The remaining 51% of IT projects are "challenged" - seriously late, over budget and lacking expected features.

Frank Hayes, 8 November 2004, Computerworld

Time and Money are Symptoms of Quality

(not vice versa)

- ? Defects get exponentially more expensive and time consuming to fix as one progresses down the System Development Lifecycle (SDLC)
- ? There needs to be a clear understanding of which 'gates' are important for ensuring quality deliverables and a successful project outcome



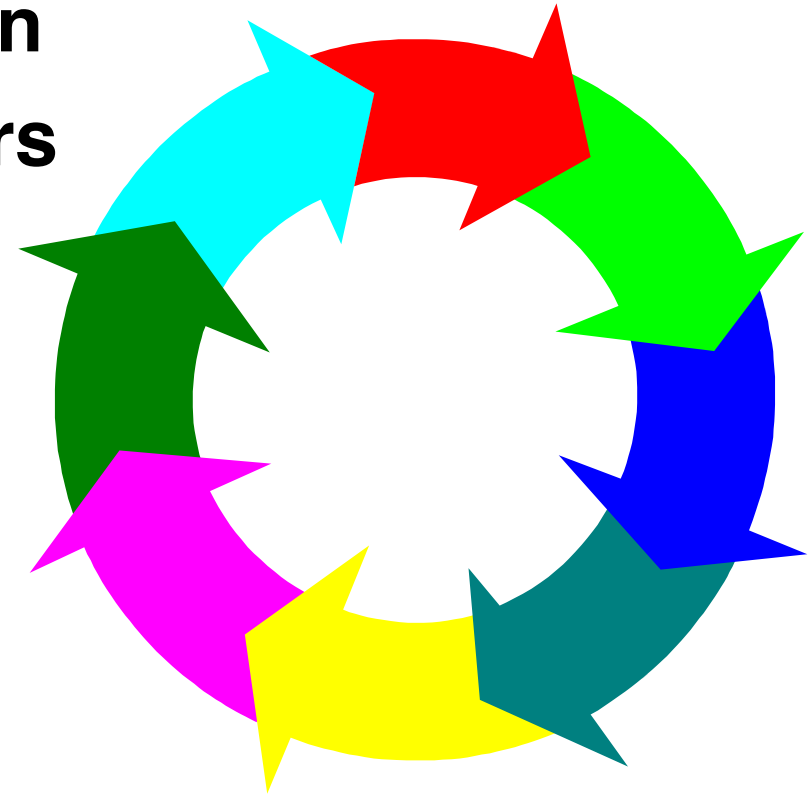
ES Project Challenges

- ❑ People Issues - Change management, capabilities of internal staff, project team mgmt, training, allocating & prioritising resources, managing & working with consultants, discipline/commitment to project
- ❑ Process Issues – project management, difficulty reengineering business process, transitioning from one stage of project to another, reaching goals & realising benefits
- ❑ Technology Issues – Software functionality, setup of reports, upgrades, preparing data, managing applications of ES

Critical Success Factors

- **Clear project aims**
- **Well-planned project organization**
- **Competent project team members**
- **Good internal communications**
- **Fast decision processes**
- **Early inclusion of end-users
(higher degree of acceptance)**
- **Early concentration on key
questions - 80:20 rule**

 **Key success factor: project leader**





How the customer
explained it



How the project leader
understood it



How the analyst
designed it



How the programmer
wrote it



What the beta testers
received



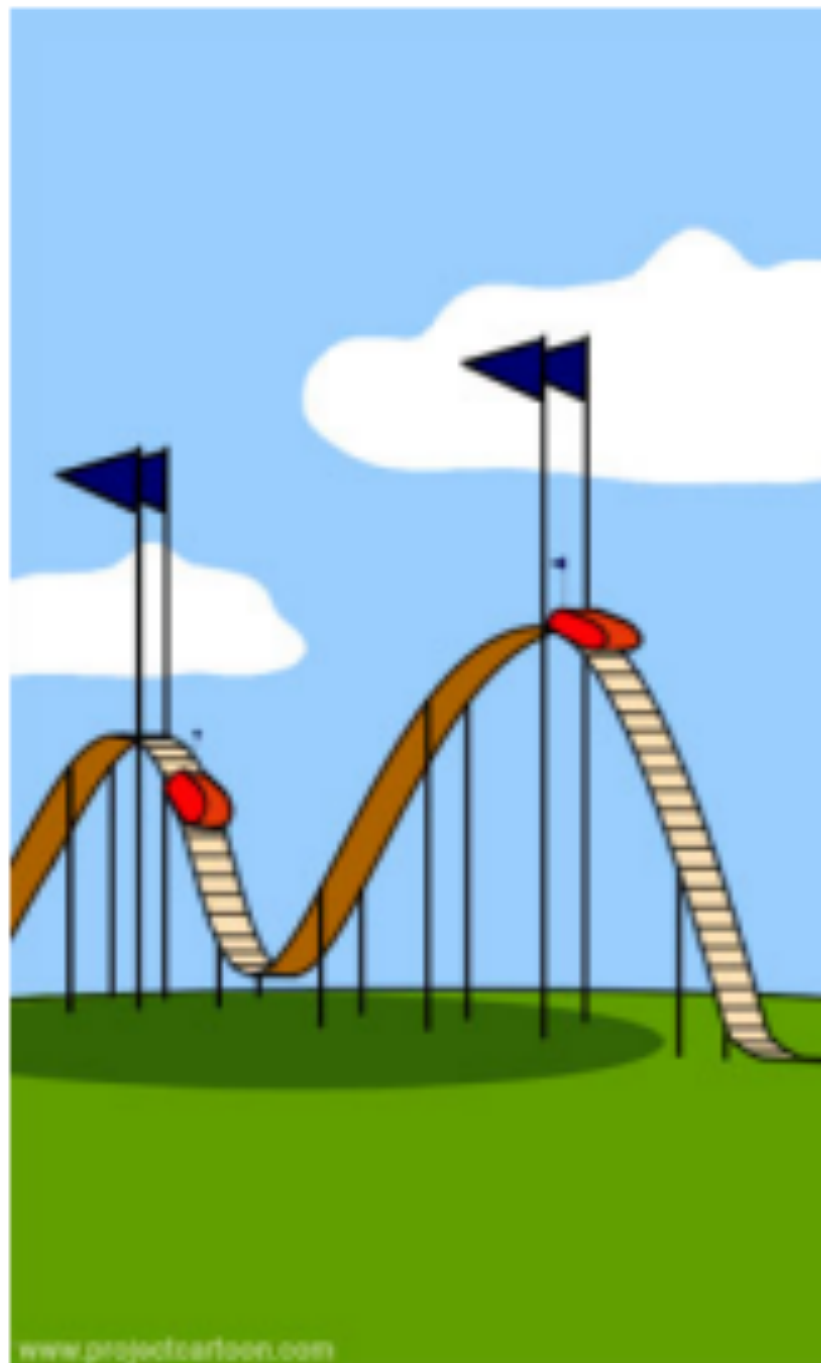
How the business
consultant described it



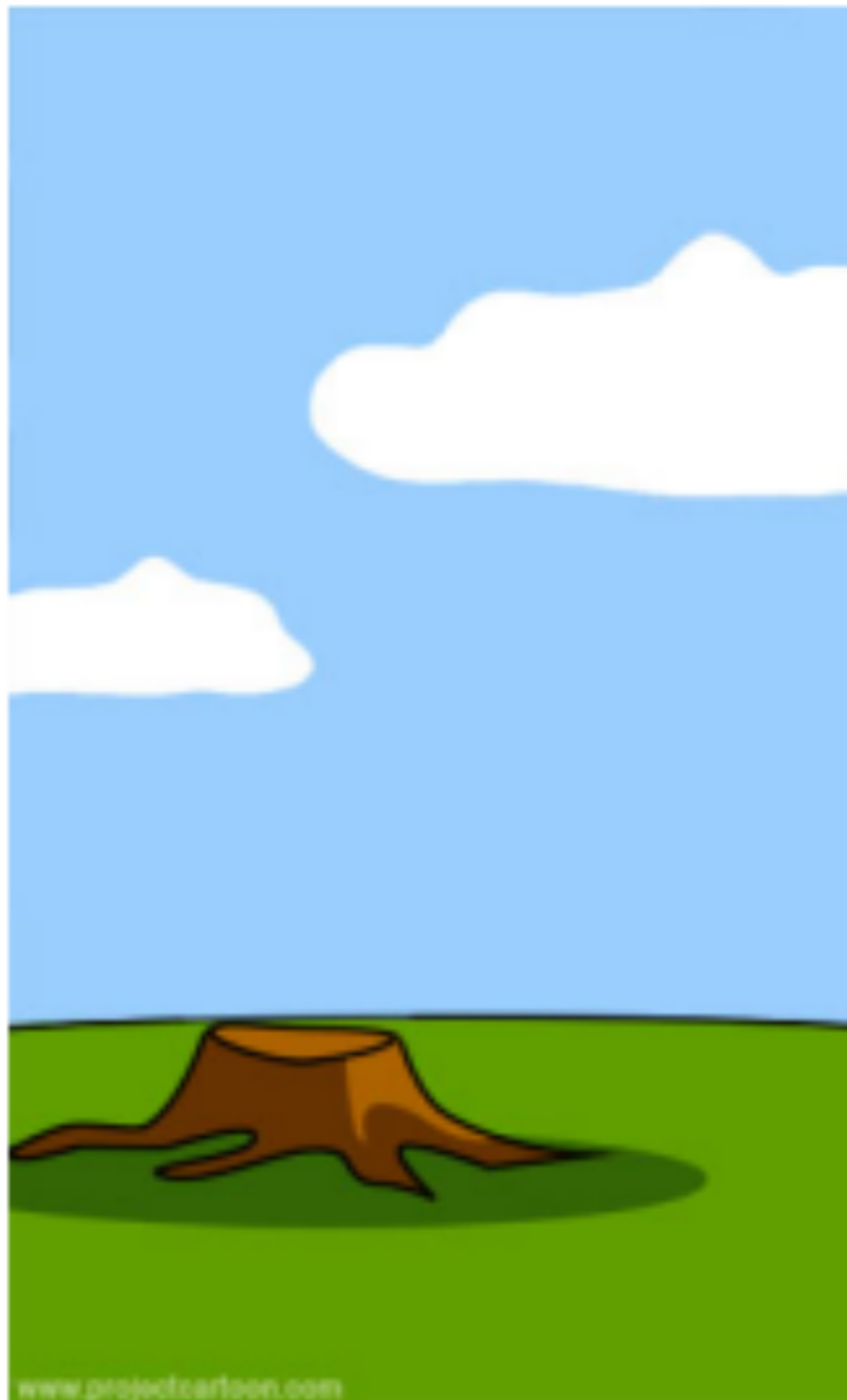
How the project was
documented



What operations
installed



How the customer was
billed



How it was supported



What marketing
advertised



What the customer
really needed

Implementation Scenarios and Strategies

Implementation Scenarios

**Jump-start
implementation**

**Lean - software
implementation**

**BPR - permanent
system improvements
(KAIZEN)**



**Two-phase
implementation**

**BUSINESS PROCESS
REENGINEERING**



**Software
implementation**

**Parallel
projects**

Business reengineering / BPR

Software implementation

**Simultaneous;
R/3-controlled
transfer**

**SOFTWARE
BASED
IDEAL
DESIGN**

BPR

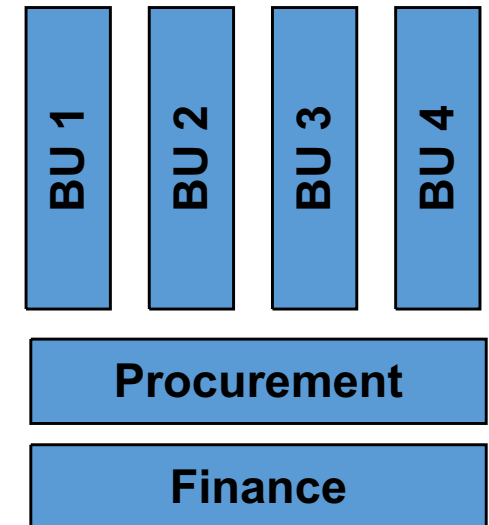
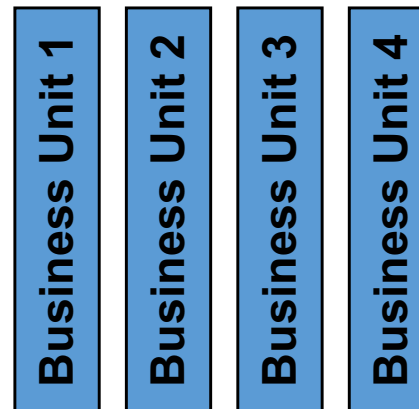
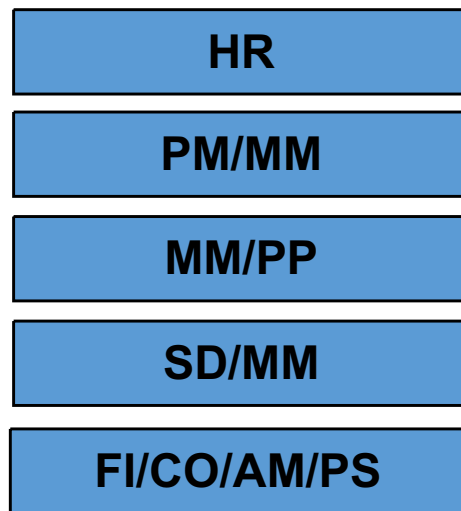
**Software
implementation**

Time



Implementation Strategy

- Horizontal
- Vertical
- Combinations



Implementation Strategy Drivers

- Number, Age and Complexity of Legacy Systems
- Cost of Temporary or Throw-Away Interfaces
- Organization's Ability to Sustain Momentum
- Desire for Integration
- Degree of Focus on Business Process Changes
- Business Benefit and Priorities

Lessons Learned

- Start with Horizontal “Foundation” Projects Like Finance, Procurement, and Engineering
- Follow Foundation Projects Closely with Smaller Business Units Using a Vertical Strategy
- Ensure Initial Business Unit Implementations are Representative of the Various Lines of Business

Implementation Strategies

- Phased
 - by business unit
 - by geographical location
 - by application
- Big-Bang
 - Enterprise wide implementation

Implementation Considerations in a Phased Approach

- Breaks ES implementation into manageable projects
- Manageable impact on individual business units
- Allows incremental learning and knowledge transfer
- Leverage knowledge and new skills acquired to later implementations
- The full benefit of common business processes won't be until all business units are implemented
- Interim interfaces to legacy systems
- Potentially extends time frame for full implementation

Implementation Considerations in a Big-Bang Approach

- Generally the quickest path to complete implementation
- Eliminates interim interfaces
- Quicker transition to new business processes for the enterprise
- Longer time period for first results
- Puts most difficult business unit on the critical path for others
- No leveraging of lessons learned

Implementation Options (Davenport, 2000)

Scope

Process Phasing	Big Bang
Incremental	Geographic or Business Unit Phasing

Function

Sequence?

Hybrid

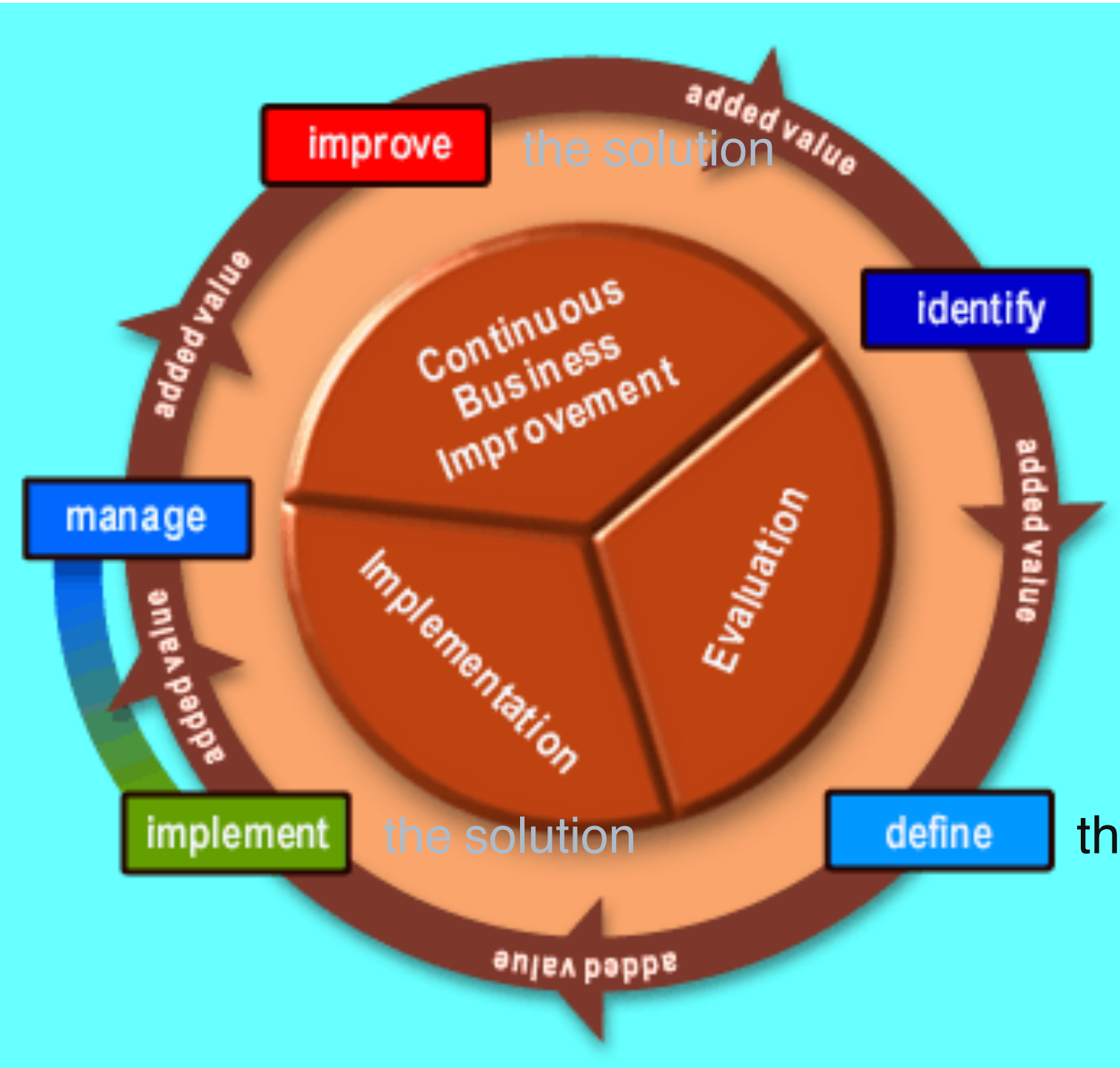
Phased but Quick Implementation

- The Big Bang approach is not advised
- Phase your implementation process – eg. implement financials and sales first and then HR and Manufacturing
- Each stage and/or phase can be very different for different companies – why?
- Callaway (1999) advises quick implementation and points to rapid implementation methodologies – why?

Implementation Phases

ValueSAP Life Cycle Approach

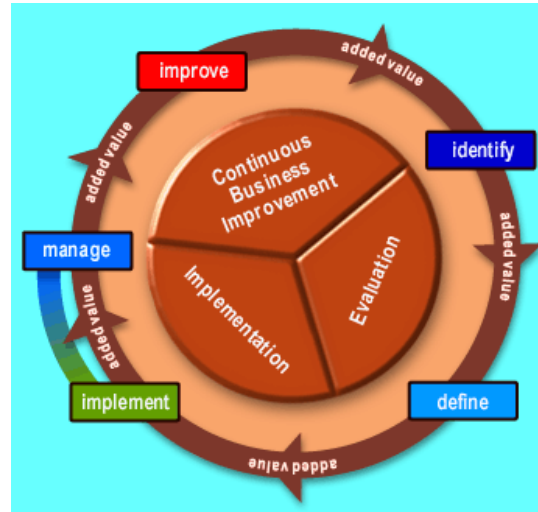
Reducing Time-
to-Benefit



Goals of the Life Cycle Approach

OPERATIONS & CONTINUOUS BUSINESS IMPROVEMENT

1. Assess baseline value for ongoing business performance
2. Measure business performance against industry best practices
3. Justify continuous business change and performance tuning
4. Realize continuous improvement



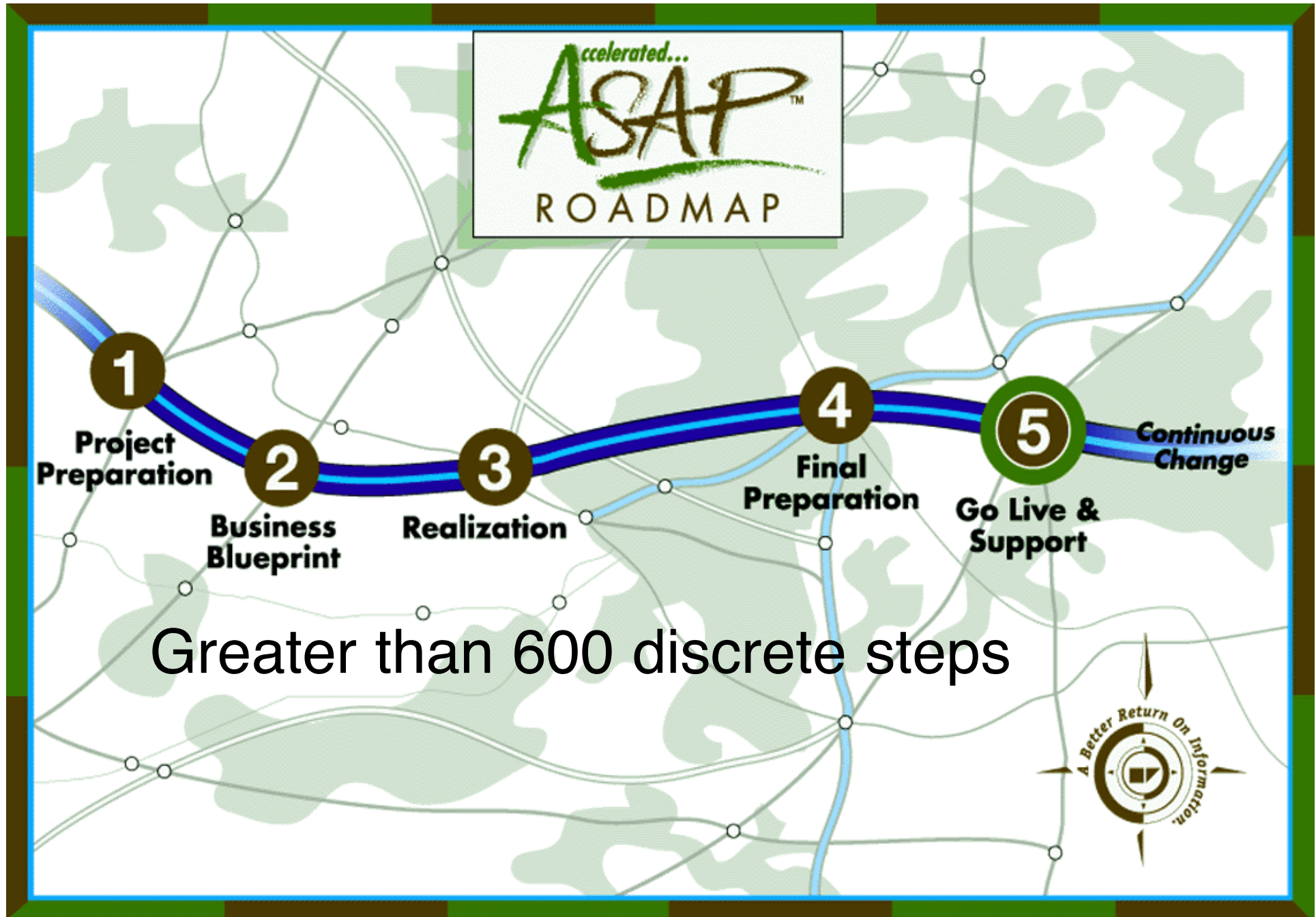
EVALUATION

1. Develop business strategy
2. Identify business solution
3. Align industry solution with key process indicators (KPIs)
4. Estimate total cost of ownership
5. Use integrated content to develop business case

IMPLEMENTATION (uses the ASAP methodology)

1. Incorporate business case as part of ASAP methodology
2. Ensure KPIs are measured throughout implementation
3. Use business case for scope/change control
4. Use integrated content to monitor project risk performance

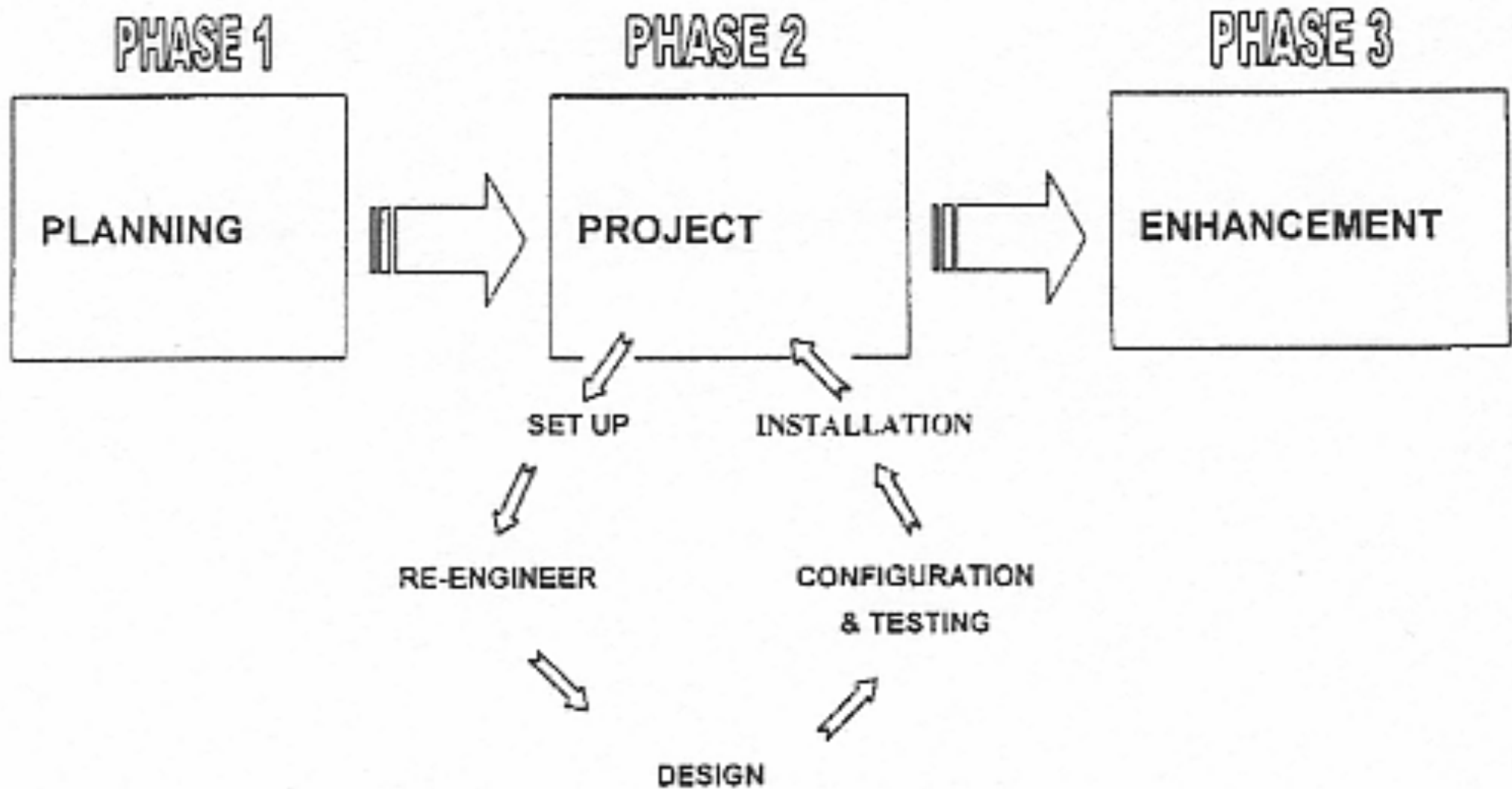
ValueSAP Implementation Phase



SOA Life cycle Phases



Phases from Project Phase Model



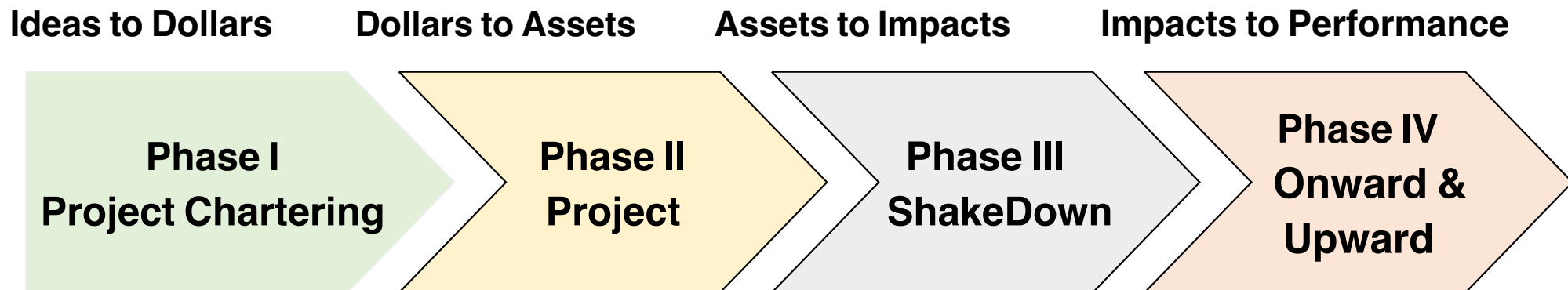
Phases from ES adopters viewpoint

The three phases of the adopter's ES experience are:

- the *Project Phase* during which ES software is configured and rolled out to the organization
- the *Shakedown Phase* during which the company makes the transition from “go live” to “normal operations”, and
- the *Onward and Upward Phase* - in which the company captures the majority of business benefits from the ES and plans the next steps for technology implementation and business improvement.

Enterprise Systems Life cycle Phases

- Chartering: Decisions defining business case and solutions constraints
- Project (configuration & Roll Out): Getting system and users “up and running”
- Shakedown: Stabilizing, eliminating “bugs”, getting to normal operations
- Onward & Upward: Maintaining system, supporting users, getting results, upgrading



Markus & Tanis (2000)

Detailed Implementation Roadmap

- Chartering
- Project
 - Project Preparation
 - Plan Business Processes
 - Build or Configure System
 - Test & Validate
 - Final Preparation
 - Go Live
- Shakedown
- Onward & Upward

Chartering

- Should my company implement an ES?
- What are the System Prerequisites?
- Who will be the decision makers?
- Vendor selection
- Implementation Partner selection
- Business Case – Cost versus Benefits

Chartering - Business Case - Costs

- Hardware - Setting up client-server infrastructure, Servers (production, backup, testing) & Tools
- Software - License: number of users, company-wide license, Support and Maintenance costs
- Costs of changing organization, Consultants & Contractors, Additional software (such as databases), Data conversion, Training, Integration, Testing, Disruption, Fall back in customer satisfaction, Loss of good people, Pre-existing infrastructure, Organizational breadth, Planned breadth, Planned redesign, Legacy systems that must be integrated, Amount of customization of system, Number of reports to create, Data Warehousing, “Rework”
- Concept of total cost of ownership throughout the systems lifecycle

Chartering - Business Case - Benefits

- New approaches of work - automation of work processes, employee self services, data rationalization, process changes, organizational changes: e.g., shared services for common processes, inventory and other costs of production savings
- Revenue enhancement - Improvement in customer service , Ease of expansion and growth, Better decision making
- Total Costs of Ownership reduction - Avoiding of ad hoc system changes, avoid legacy system upgrade and support costs, in house versus packaged
- Improve visibility into supply chain, tightly integrated business processes, Easy to use systems, basic infrastructure for pursuing SCM/CRM/SEM.

Project Preparation

- Resources required to complete project within budget and on time
- Project team formation including hiring of consultants, contractors - composed of Technical and Business people
- Executive Sponsor - Relate system to overall strategy, Communicate value/importance of project to org, Develop & advertise performance improvement objectives, Create necessary org changes, ensure project proceeds on time
- Project Leader/Manager - Combination of technologist, business expert, drill sergeant, motivational speaker, politician, and psychologist
- Identify Business Process Owners, Super Users, Vision and Planning team, and Implementation team

Plan Business Processes

- Companies examine their current business process, industry best practices, & the Enterprise Systems process model & combine them to design a new business process model
- Leverage the business process model as a starting point for maximum automation of systems & configuration customising.
- Process identification, Process modelling (AS-IS), Process analysis and Process improvement (TO-BE)
- Examine the Purpose, Place, Sequence, Person, Means with a view to SCAMPER (Substitute/Simplify, Combine, Adapt, Modify/Magnify, Purpose, Eliminate/Minimize and Rearrange/Reverse)
- Based on the strategy and TO-BE processes define requirements

Requirements - FURPS

A need that must be fulfilled by a system or a product. They define what the system should do or how it should behave to satisfy the needs of its users, stakeholders, and the business as a whole.

Functional: functions, capabilities, and behaviours

Non-Functional: performance, reliability, security, usability, scalability, maintainability, and compliance with standards or regulations.

Business: high-level objectives or needs that aligns with the organization's goals, strategies, and priorities.

User: needs, preferences, and expectations.

System: capabilities and constraints of the hardware, software, and infrastructure components.

Design: data structures, algorithms, interfaces, and software architecture patterns.

Building or Configuring the System

Tailoring Type	Description
Configuration	Setting of parameters (or tables), in order to choose between different executions of pro-cesses and functions in the software package
Screen masks	Creating of new screen masks for input and output (soft copy) of data
Workflow programming	Creating of non-standard workflows
Extended reporting	Programming of extended data output and reporting options
User exits	Programming of additional software code in an open interface
Programming	Programming of additional applications, without changing the source code (using the computer language of the vendor)
Interface development	Interfaces to legacy systems or 3rd party products
Package code modification	Changing the source-codes ranging from small change to change whole modules

Testing and Validating the System

- Ensure that system works correctly from a Technical as well as practical perspective to avoid catastrophes – FURPS.
- Testing occurs through the project, ongoing process concurrent with designing and configuring of system
- Steps: Planning, setting up test env, testing interfaces, availability, performance, user-acceptance testing, training power users, & developing training materials.
- Technical testing: concurrent, combinations of transactions, distributed architecture, interfacing
- Testing Tools: Tests performance, unattended test execution, automated documentation, and analysis
- Testing Business Processes: UAT, VIP, conference room pilots, unit, module, scenario, integration testing

Final Preparation

- Resolve any problems with project – problems that appeared during testing, resolve, and retest
- Final validation of all interfaces
- Ensure that data to be imported into system is in the correct format and ready to use
- Train all employees

Go Live

- Two major steps
 - Activating the new system and
 - Transitioning from the old apps to the new ones
- Use weekends/holidays to go live – minimal impact on business, also allows you to check how the system performs
- Clean cut-over – Stop Old Sys – Start New ES
- Parallel till kinks are ironed out and New ES is working according to specs
- Parallel and slowly phasing out older sys
- Phased approach – Transition to New ES in smaller, representative Business Unit – Learn from experience and then transition to New ES in other Units.....

